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Inventor(s)

NORIMASA; Oga et al.

PERIPHERAL LIGHT-EMITTING LINEAR LIGHT GUIDE MEMBER AND METHOD OF MANUFACTURING THE SAME

Abstract

A peripheral light-emitting linear light guide member having low manufacturing cost and high uniformity of light intensity is achieved. A peripheral light-emitting optical fiber **3** according to one embodiment includes: an optical fiber **4** including a core **41** exposed from a cladding **42** at one end portion **44**; and a light-scattering resin film **5** including a light-transmittable base material **50** and light-scattering particles **51** dispersed in the base material **50**, and the light-scattering resin film **5** includes a first region **61**, a second region **62**, and a third region **63** covering an outer periphery surface of the end portion **44** of the core **41**. A minimum film thickness of the second region **62** is equal to or larger than a maximum film thickness of the first region **61**, a minimum film thickness of the third region **63** is equal to larger than a maximum film thickness of the second region **62**. Each film thickness of the first region **61** and the second region **62** gradually increases in a direction toward the tip end side of the end portion **44**, and reaches the maximum film thickness of each region, and a film thickness of the third region **63** gradually increases in the direction toward the tip end side of the end portion **44**, and reaches a maximum film thickness of the third region **63**, and then, gradually decreases to a predetermined film thickness.

Inventors: NORIMASA; Oga (Tokyo, JP), TAKAHASHI; Ryuta (Tokyo, JP)

Applicant: Proterial, Ltd. (Tokyo, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The application claims foreign priority benefits under 35 U.S.C. § 119 to Japanese Patent Application No. 2024-30760 filed on Feb. 29, 2024, the content of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD OF THE INVENTION

[0002] The present invention relates to a peripheral light-emitting linear light guide member and a method of manufacturing the same.

BACKGROUND OF THE INVENTION

[0003] A catheter treatment has been known, the catheter treatment being performed by inserting an optical fiber catheter equipped with an optical fiber into a human body's lumen organs (such as esophagus or bowel), blood vessel or heart and treating an affected area by light emitted from a core of the optical fiber. As a peripheral light-emitting linear light guide member for use in such catheter treatment, the Applicant of the present application has suggested a peripheral light-emitting linear light guide member described in Japanese Patent Application Laid-open Publication No. 2022-158714 (Patent Document 1).

[0004] The peripheral light-emitting linear light guide member described in the above-described Patent Document includes: an optical fiber with a core exposed by removing a cladding; and a light-scattering member covering an outer periphery surface of the exposed core. The light-scattering member is made of a light-transmittable base material with a refractive index higher than that of the core and light-scattering particles dispersion-mixed with the base material.

[0005] In order to enhance uniformity of light intensity in an axial direction, the light-scattering member includes a plurality of layers each with a different ratio of the mixing of the light-scattering particles to the base material, and at least part of each layer overlaps in a radial direction of the core.

[0006] In order to form the light-scattering member as described above, solutions of several types each with a different ration of the mixing of the light-scattering particles are prepared, and the solutions are sequentially adhered to the outer circumference of the core, and are cured.

SUMMARY OF THE INVENTION

[0007] The optical fiber catheter for use in catheter treatment is disposable, and its low cost performance is desired. However, in the peripheral light-emitting linear light guide member including the light-scattering member formed as described above, many manhours and long time are required to form the light-scattering member. Therefore, it is not easy to achieve the low cost performance.

[0008] A peripheral light-emitting linear light guide member according to one embodiment includes: an optical fiber including a cladding and a core exposed from the cladding at one end portion; and a light-scattering resin film including a light-transmittable base material with a refractive index higher than that of the core and light-scattering particles dispersed in the base material. The light-scattering resin film includes a first region, a second region, and a third region covering an outer periphery surface of the end portion of the core, and the first region, the second region, and the third region are aligned in this order from a base end side to a tip end side of the

end portion. A minimum film thickness of the second region is equal to or larger than a maximum film thickness of the first region, and a minimum film thickness of the third region is equal to larger than a maximum film thickness of the second region. Each film thickness of the first region and the second region gradually increases in a direction toward the tip end side of the end portion, and reaches the maximum film thickness of each region. A film thickness of the third region gradually increases in the direction toward the tip end side of the end portion, and reaches a maximum film thickness of the third region, and then, gradually decreases to a predetermined film thickness.

[0009] A method of manufacturing a peripheral light-emitting linear light guide member according to one embodiment is a method of manufacturing a peripheral light-emitting linear light guide member including: an optical fiber including a core exposed from a cladding at one end portion; and a light-scattering resin film including a first region, a second region, and a third region covering an outer periphery surface of the end portion of the core. The manufacturing method includes: an optical-fiber processing step of exposing the end portion of the core from the cladding; a film-forming material preparing step of preparing a solution containing light-scattering particles dispersed in a light-transmittable solvent having a refractive index higher than that of the core; an immersing step of immersing the end portion of the core into the solution; and a pull-up step of pulling up the end portion of the core from the solution. The pull-up step includes at least a first step of forming the first region, a second step of forming the second region, and a third step of forming the third region, and the first step, the second step, and the third step are performed in this order. A pull-up start speed in the second step is equal to or higher than a pull-up maximum speed in the first step, and a pull-up start speed in the third step is equal to or higher than a pull-up maximum speed in the second step. A pull-up speed in the first step is gradually increased from a pull-up start speed to the pull-up maximum speed in the first step. The pull-up speed in the second step is gradually increased from the pull-up start speed to the pull-up maximum speed in the second step. The pull-up speed in the third step is gradually increased from the pull-up start speed to a pull-up maximum speed in the third step, and then, is gradually decreased to a predetermined speed.

[0010] According to the present invention, a peripheral light-emitting linear light guide member with high uniformity of light intensity can be manufactured at low cost.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0011] FIG. 1 is a schematic diagram showing a treatment device using a peripheral light-emitting optical fiber as a catheter and a patient as a treatment target.

[0012] FIG. 2 is a schematic diagram showing one end portion (tip end portion) of the peripheral light-emitting optical fiber inserted into the body of the patient.

[0013] FIG. 3 is a perspective view showing the outer appearance of the tip end portion of the peripheral light-emitting optical fiber.

[0014] FIG. 4 is a cross-sectional view showing the sectional structure of the tip end portion of the peripheral light-emitting optical fiber.

[0015] FIG. 5 is an explanatory diagram showing a film thickness and a length of each region of a light-scattering resin film.

[0016] FIG. 6A is an explanatory diagram showing an optical-fiber processing step.

[0017] FIG. 6B is another explanatory diagram showing the optical-fiber processing step.

[0018] FIG. 6C is still another explanatory diagram showing the optical-fiber processing step.

[0019] FIG. 7 is an explanatory diagram showing a film-forming material preparing step.

[0020] FIG. 8 is an explanatory diagram showing an immersing step.

[0021] FIG. **9A** is an explanatory diagram showing a pull-up step.

[0022] FIG. **9B** is another explanatory diagram showing the pull-up step.

[0023] FIG. **9C** is still another explanatory diagram showing the pull-up step.

[0024] FIG. **9D** is still another explanatory diagram showing a pull-up step.

[0025] FIG. **10** is a graph showing a pull-up speed in a first step, a second step, and a third step.

[0026] FIG. **11** is a graph showing measurement results of light-emission intensity of the peripheral light-emitting optical fiber.

DESCRIPTIONS OF THE PREFERRED EMBODIMENTS

[0027] Hereinafter, an example of embodiments of the present invention will be described in detail with reference to the accompanying drawings. Note that the same or substantially same configurations and components are denoted by the same reference symbols throughout all the drawings for describing the embodiments. And, the already-described configurations and components will not be repeatedly described in principle.

<General Description of Treatment Device>

[0028] FIG. **1** is a schematic diagram showing a treatment device **1** using a peripheral light-emitting linear light guide member according to the present embodiment as a catheter together with a patient “P” being treated. The treatment device **1** includes a main body **2** and a peripheral light-emitting linear light guide member **3**, and a tip end (distal end) of the peripheral light-emitting linear light guide member **3** is inserted into the patient P. The main body **2** includes a light source **21** that emits laser light. The laser light output from the light source **21** included in the main body **2** is incident from a base end (proximal end) of the peripheral light-emitting linear light guide member **3** to the peripheral light-emitting linear light guide member **3**. In the following description, the peripheral light-emitting linear light guide member **3** may be referred to as “peripheral light-emitting optical fiber **3**”

<Structure of Peripheral Light-Emitting Linear Light Guide Member>

[0029] FIG. **2** is a schematic diagram showing one end portion (tip end portion) of the peripheral light-emitting optical fiber **3** being inserted into the body of the patient P. FIG. **2** shows that the blood vessel P1 of the patient P is partially cut out, and that the tip end of the peripheral light-emitting optical fiber **3** is inserted into a blood vessel P1.

[0030] The scattered laser light “Lr” emitted from the tip end of the peripheral light-emitting optical fiber **3** irradiates a treated site P2, and activates the drugs previously contained in the treated site P2. This provides intra-vascular laser treatment.

[0031] FIG. **3** is a perspective view showing an outer appearance of the tip end of the peripheral light-emitting optical fiber **3**. FIG. **4** is a cross-sectional view showing a cross-sectional structure of the tip end of the peripheral light-emitting optical fiber **3**. The peripheral light-emitting optical fiber **3** includes: an optical fiber **4**, which directs the incident laser light to the treated site P2; and a light-scattering resin film **5** located at one end (tip end) of the optical fiber **4**.

[0032] Note that FIG. **3** and FIG. **4** exaggeratedly show the film thickness of the light-scattering resin film **5** for convenience of description.

<<Optical Fiber>>

[0033] The optical fiber **4** includes a core **41**, a cladding **42**, and a sheath **43**. At the tip end of the optical fiber **4**, the cladding **42** is exposed from the sheath **43**, and furthermore, the core **41** is exposed from the cladding **42**.

[0034] That is, the optical fiber **4** includes the core **41** exposed from the cladding **42** at one end portion. In the following description, the one end portion of the core **41** exposed from the cladding **42** may be referred to as “exposed end portion **44**” as distinguished from the other portions of the core **41**. Also, the outer periphery surface of the exposed end portion **44** may be referred to as “outer periphery surface **44a**”, and the end surface of the exposed end portion **44** may be referred to as “tip end surface **44b**”.

[0035] The core **41** of the optical fiber **4** is made of quartz glass, and the cladding **42** is made of a

polymer. That is, the optical fiber **4** is a quartz-glass optical fiber. Also, the sheath **43** of the optical fiber **4** is made of fluorine-based resin. More specifically, the sheath **43** is made of ETFE (ethylene-tetrafluoroethylene polymer).

[0036] The core **41** has a diameter **D1** of $200\text{ mm}\pm 5\text{ }\mu\text{m}$, the cladding **42** has a diameter **D2** of $225\text{ mm}\pm 5\text{ }\mu\text{m}$, and the sheath **43** has a diameter **D3** of $500\text{ mm}\pm 30\text{ }\mu\text{m}$. The refractive index of the core **41** is higher than the refractive index of the cladding **42**. Thus, the incident light on the core **41** propagates through the core **41** while being totally reflected at an interface between the core **41** and the cladding **42**.

<<Light-Scattering Resin Film>>

[0037] The light-scattering resin film **5** covers a predetermined range of the exposed end portion **44**. Note that a root of the exposed end portion **44** has a non-covered portion **44c** neither covered with the cladding **42** nor the light-scattering resin film **5**.

[0038] The light-scattering resin film **5** scatters and emits light emitted from surfaces (the outer periphery surface **44a** and the tip end surface **44b**) of the exposed end portion **44**. The light-scattering resin film **5** includes a light-transmittable base material **50** with a refractive index higher than that of the core **41** and many light-scattering particles **51** that scatter the incident light on the base material **50**.

[0039] The base material **50** is made of thermosetting resin. In the present embodiment, silicone resin is used. The refractive index of silicone resin is, for example, 1.52. On the other hand, the refractive index of the core **41** is, for example, 1.46.

[0040] The light-scattering particles **51** are metal particles that reflect the incident light on the base material **50**. In the present embodiment, rutile-type titanium oxide is used. However, the light-scattering particles **51** are not limited to be made of titanium oxide. For example, fine metal powder of aluminum oxide (alumina), silver, copper, iron, or an alloy thereof may be used as the light-scattering particles **51**.

[0041] The light-scattering particles **51** are dispersion-mixed at a constant ratio over the entire base material **50**. The phrase “being dispersion-mixed at a constant ratio” described here means that the light-scattering particles **51** are not unevenly distributed in part of the base material **50** but are mixed so as to be evenly dispersed over the entire base material **50**. Note that the light-scattering particles **51** are too small to be recognized by the naked eyes. However, in the attached drawings of the specification, the sizes of the light-scattering particles **51** are exaggeratedly shown.

<<Each Region of Light-Scattering Resin Film>>

[0042] The light-scattering resin film **5** includes a first region **61**, a second region **62**, a third region **63**, a fourth region **64**, and a fifth region **65**. The first region **61**, the second region **62**, the third region **63**, the fourth region **64**, and the fifth region **65** are aligned in this order from the base end side (root side) toward the tip end side of the exposed end portion **44**. Also, the first region **61**, the second region **62**, the third region **63**, and the fourth region **64** cover the entire outer circumference of the outer periphery surface **44a** of the exposed end portion **44**, and the fifth region **65** covers the entire tip end surface **44b** of the exposed end portion **44**.

[0043] The film thicknesses of the light-scattering resin films **5** in the first region **61**, the second region **62** and the third region **63** are gradually larger in this order. On the other hand, the film thickness of the light-scattering resin film **5** in the fourth region **64** is substantially constant, and is smaller than a minimum film thicknesses of the same in all of the first region **61**, the second region **62** and the third region **63**. Also, the fifth region **65** of the light-scattering resin film **5** has a substantially semispherical shape.

[0044] Note that the film thickness of the light-scattering resin film **5** is the film thickness of the light-scattering resin film **5** in a direction perpendicular to a center axis line CA of the core **41**.

[0045] In another viewpoint, each of the first region **61**, the second region **62**, and the third region **63** of the light-scattering resin film **5** is a “gradually-increasing portion” having a thickness gradually increasing toward the tip end side of the exposed end portion **44**. The fourth region **64** of

the light-scattering resin film 5 is an “annular small-thickness portion” or “cylindrical small-thickness portion” provided closer to the tip end side than the gradually-increasing portion. The fifth region 65 of the light-scattering resin film 5 is a “tip end-covering portion” provided closer to the tip end side than the “annular small-thickness portion” or “cylindrical small-thickness portion”

<<Length of Each Region>>

[0046] FIG. 5 is an explanatory diagram showing the film thickness and the length of each region of the light-scattering resin film 5. The “length” in the following description means a length along the center axis line CA of the core 41 (length in the axial direction) unless otherwise specified.

[0047] A length CL of the exposed end portion 44 of the core 41 is 58.0 mm, and a length L of the light-scattering resin film 5 from a start edge of the first region 61 to an end edge of the third region 63 is 54.0 mm. More specifically, a length L1 of the light-scattering resin film 5 in the first region 61 is 20.0 mm, a length L2 of the same in the second region 62 is 20.0 mm, and a length L3 of the same in the third region 63 is 14.0 mm. Also, a length L4 of the light-scattering resin film 5 in the fourth region 64 is 2.0 mm, and a length L0 of the non-covered portion 44c of the core 41 is 2.0 mm.

[0048] As described above, the length L2 of the second region 62 is equal to or larger than the length L1 of the first region 61 ($L2 \geq L1$). More specifically, the length L2 of the second region 62 is equal to the length L1 of the first region 61 ($L2 = L1$).

[0049] Also, the length L3 of the third region 63 is smaller than the length L1 of the first region 61, and smaller than the length L2 of the second region 62 ($L1, L2 > L3$).

[0050] As described above, the light-scattering resin film 5 of the present embodiment satisfies a relation “ $L1 \geq L2 \geq L3$ ”, and also satisfies a relation “ $L1, L2, L3 > L4$ ”.

<<Film Thickness of First Region>>

[0051] The film thickness of the first region 61 gradually increases in a direction toward the tip end side of the exposed end portion 44 to reach a maximum film thickness T1max. As a result, the surface of the first region 61 is tilted with respect to the center axis line CA of the core 41. In other words, the surface of the first region 61 is a tapered surface.

<<Film Thickness of Second Region>>

[0052] The film thickness of the second region 62 gradually increases in the direction toward the tip end side of the exposed end portion 44 to change from a minimum film thickness T2min to a maximum film thickness T2max. As a result, as similar to the surface of the first region 61, the surface of the second region 62 is tilted with respect to the center axis line CA of the core 41. In other words, as similar to the surface of the first region 61, the surface of the second region 62 is a tapered surface.

[0053] The minimum film thickness T2min of the second region 62 is equal to or larger than the maximum film thickness T1max of the first region 61 ($T2min \geq T1max$). More specifically, the minimum film thickness T2min of the second region 62 is equal to the maximum film thickness T1max of the first region 61 ($T2min = T1max$).

[0054] Also, the maximum film thickness T2max of the second region 62 is larger than the maximum film thickness T1max of the first region 61 ($T2max > T1max$).

<<Film Thickness of Third Region>>

[0055] The film thickness of the third region 63 gradually increases in the direction toward the tip end side of the exposed end portion 44 to change from a minimum film thickness T3min to a maximum film thickness T3max. Furthermore, after reaching the maximum film thickness T3max, the film thickness of the third region 63 gradually decreases to a predetermined film thickness T3mid in the direction toward the tip end side of the exposed end portion 44. As a result, the surface of the third region 63 is tilted with respect to the center axis line CA of the core 41. Also, the surface of the third region 63 is a substantially tapered surface as a whole.

[0056] The minimum film thickness T3min of the third region 63 is equal to or larger than the maximum film thickness T2max of the second region 62 ($T3min \geq T2max$). More specifically, the

minimum film thickness T3min of the third region **63** is equal to the maximum film thickness T2max of the second region **62** (T3min=T2max).

[0057] Also, the maximum film thickness T3max of the third region **63** is larger than the maximum film thickness T2max of the second region **62** (T3max>T2max). Furthermore, the maximum film thickness T3max is a maximum value of the film thickness of the light-scattering resin film **5**. That is, a maximum film thickness portion of the light-scattering resin film **5** is in the third region **63**.

[0058] Furthermore, the predetermined film thickness T3mid is smaller than the maximum film thickness T3max of the third region **63**, and is larger than the minimum film thickness T3min (T3min<T3mid<T3max).

<<Film Thickness of Fourth Region>>

[0059] The film thickness of the fourth region **64** is smaller than the film thickness of the first region **61** in the entire fourth region **64**. Furthermore, as described above, the film thickness of the fourth region **64** is substantially constant in the entire fourth region **64**. As a result, the surface of the fourth region **64** is parallel to the center axis line CA of the core **41**.

[0060] As described above, the light-scattering resin film **5** in the present embodiment satisfies a relation “T1max≤T2min<T2max≤T3min<T3mid<T3max”.

[0061] Since the film thickness of the light-scattering resin film **5** changes along the axial direction of the exposed end portion **44** as described above, an annular step surface **66** parallel to the tip end surface **44b** of the exposed end portion **44** is provided between the third region **63** and the fourth region **64**.

<Manufacturing Method>

[0062] Next, one example of a method of manufacturing the peripheral light-emitting optical fiber **3** will be described. The manufacturing method according to the present embodiment includes at least an optical-fiber processing step, a film-forming material preparing step, an immersing step, and a pull-up step.

[0063] Furthermore, the pull-up step includes at least a first step, a second step, a third step, a fourth step, and a fifth step. The first step, the second step, the third step, the fourth step, and the fifth step are performed in this order.

<<Optical-Fiber Processing Step>>

[0064] FIGS. **6A**, **6B** and **6C** are explanatory diagrams each showing the optical-fiber processing step. In the optical-fiber processing step, the optical fiber **4** including the core **41**, the cladding **42**, and the sheath **43** is prepared as shown in FIG. **6A**.

[0065] Next, as shown in FIG. **6B**, the sheath **43** is removed over any length range, and the cladding **42** is exposed. In the present embodiment, the sheath **43** is removed over 75.0 mm.

[0066] Then, as shown in FIG. **6C**, the cladding **42** is removed to expose the end portion of the core **41** from the cladding **42**, and the exposed end portion of the core **41** is cut out so as to have any length.

[0067] More specifically, the exposed end portion of the core **41** is cut out so as to have any length longer than an intended length of the light-scattering resin film **5**. In the present embodiment, an unnecessary portion (extra length portion) is cut out so that the end portion of the core **41** has a length of 58.0 mm.

[0068] By the optical-fiber processing step as described above, the optical fiber **4** having the core **41** including the exposed end portion **44** is obtained.

[0069] Note that the cladding **42** can be removed by using, for example, an organic solvent such as acetone. Also, the unnecessary portion of the core **41** can be broken by, for example, making a scratch on the core **41** by using a sharp jig T, and then, applying stress thereon.

<<Film-Forming Material Preparing Step>>

[0070] FIG. **7** is an explanatory diagram showing the film-forming material preparing step. In the film-forming material preparing step, the light-scattering particles **51** are dispersion-mixed with a light-transmittable solvent **71** having a refractive index higher than that of the core **41**. More

specifically, a predetermined amount (for example, 1 mg/mL) of the light-scattering particles **51** is added and dispersed into the liquid solvent **71** contained in a container **72**.

[0071] As a result, solution **70** (FIG. **8**) in which the light-scattering particles **51** of a predetermined ratio are dispersion-mixed with the solvent **71** is obtained. That is, a film-forming material is obtained.

[0072] In the present embodiment, note that silicone resin (refractive index is 1.52) is used as the solvent **71**. Also, rutile-type titanium oxide is used as the light-scattering particles **51**. In the film-forming material preparing step, the viscosity of the solution **70** may be adjusted by adding an organic solvent such as toluene or acetone.

<<Immersing Step>>

[0073] FIG. **8** is an explanatory diagram showing the immersing step. In the immersing step, the exposed end portion **44** of the core **41** is immersed into the solution **70** prepared in the film-forming material preparing step. More specifically, a predetermined length range of the exposed end portion **44** of the core **41** is vertically immersed into the solution **70**.

[0074] In the present embodiment, the exposed end portion **44** of the core **41** is immersed down to a position of 2.0 mm in the solution **70** from the end surface of the cladding **42**. In other words, the exposed end portion **44** of the core **41** is immersed down to a position of 56.0 mm in the solution **70** from the tip end surface **44b**.

<<Pull-Up Step>>

[0075] FIGS. **9A**, **9B**, **9C** and **9D** are explanatory diagrams each showing the pull-up step. More specifically, FIG. **9A** is an explanatory diagram showing a first step of the pull-up step. FIGS. **9B** and **9C** are explanatory diagrams showing a second step and a third step, respectively. FIG. **9D** is an explanatory diagram showing a state of the optical fiber **4** immediately after end of the pull-up step.

[0076] In the pull-up step, the exposed end portion **44** of the core **41** is vertically pulled up at a predetermined speed from the solution **70**, thereby forming the light-scattering resin film **5**. That is, by a dipcoat film-forming method, the light-scattering resin film **5** having a predetermined film thickness and cross-sectional shape is formed. More specifically, the light-scattering resin film **5** including the first region **61** to the fifth region **65** shown in FIGS. **4** and **5** is formed.

[0077] In the pull-up step using the dipcoat film-forming method, by controlling the pull-up speed for the exposed end portion **44**, a thickness (height) of the solution **70** (film thickness of coating) to be adhered to the surface of the exposed end portion **44** can be controlled.

[0078] That is, by controlling the pull-up speed for the exposed end portion **44**, the film thickness of each region of the light-scattering resin film **5** can be controlled. In the present step, the thickness of the solution **70** to be adhered to the surface of the exposed end portion **44** can be calculated from the following equation (1).

[00001][Equation1]
$$h = \frac{0.94(\eta U)^{2/3}}{1/6 \sqrt{g}} \quad \text{Equation1}$$

[0079] Here, terms in Equation 1 represent “h”: the thickness (m) of the solution **70** to be adhered to the surface of the exposed end portion **44**, “η”: viscosity (Pa.Math.s) of the solution **70**, “U”: pull-up speed (m/s) for the exposed end portion **44**, “γ”: surface tension (mN/m) of the solution **70**, “ρ”: density (kg/m.sup.3) of the solution **70**, “g”: gravitational acceleration (m/s.sup.2).

[0080] From Equation 1 above, it can be found that the thickness of the solution **70** to be adhered to the surface of the exposed end portion **44** is larger as the pull-up speed for the exposed end portion **44** is higher while the thickness of the solution **70** to be adhered to the surface of the exposed end portion **44** is smaller as the pull-up speed for the exposed end portion **44** is lower.

[0081] Therefore, by increasing or decreasing the pull-up speed, the first region **61** to the fourth region **64** each having the above-described film thickness can be formed.

[0082] FIG. **10** is a graph showing each pull-up speed in the first step (FIG. **9A**), a second step (FIG. **9B**), and a third step (FIG. **9C**).

First Step

[0083] The first step shown in FIG. 9A is a step of forming the first region **61** shown in FIGS. 4 and 5. As shown in FIG. 5, the film thickness of the first region **61** gradually increases in the direction toward the tip end side of the exposed end portion **44** to reach the maximum film thickness $T1_{max}$.

[0084] Thus, in the first step, the pull-up speed gradually increases from a pull-up start speed $V1$ in the first step to a pull-up maximum speed $V2$ in the first step.

[0085] More specifically, in the first step, the exposed end portion **44** of the core **41** is vertically pulled up from the solution **70** at a pull-up speed satisfying a cubic equation (“ $speed=ax\cdot sup.3+bx\cdot sup.2+cx+d$ ” (terms “ a ”, “ b ”, “ c ”, and “ d ” are rational numbers)).

Second Step

[0086] The second step of the pull-up step shown in FIG. 9B is a step of forming the second region **62** shown in FIGS. 4 and 5. As shown in FIG. 5, the film thickness of the second region **62** gradually increases in the direction toward the tip end side of the exposed end portion **44** to change from the minimum film thickness $T2_{min}$ to the maximum film thickness $T2_{max}$.

[0087] Also, the minimum film thickness $T2_{min}$ of the second region **62** is equal to the maximum film thickness $T1_{max}$ of the first region **61** ($T2_{min}=T1_{max}$). Furthermore, the maximum film thickness $T2_{max}$ of the second region **62** is larger than the maximum film thickness $T1_{max}$ of the first region **61** ($T2_{max}>T1_{max}$).

[0088] Thus, in the second step, the pull-up speed that is gradually increased to the pull-up maximum speed $V2$ in the first step is continuously gradually increased to a pull-up maximum speed $V3$ in the second step.

[0089] More specifically, in the second step, the exposed end portion **44** of the core **41** is vertically pulled up from the solution **70** at a pull-up speed satisfying a cubic equation ($speed=ex\cdot sup.3+fx\cdot sup.2+gx+h$ (terms “ e ”, “ f ”, “ g ”, and “ h ” are rational numbers)).

[0090] As described above, the pull-up maximum speed $V2$ in the first step is also a pull-up start speed in the second step. In other words, the pull-up start speed in the second step is equal to or higher than the pull-up maximum speed $V2$ in the first step.

Third Step

[0091] The third step of the pull-up step shown in FIG. 9C is a step of forming the third region **63** shown in FIGS. 4 and 5. As shown in FIG. 5, the film thickness of the third region **63** gradually increases in the direction toward the tip end side of the exposed end portion **44** to change from the minimum film thickness $T3_{min}$ to the maximum film thickness $T3_{max}$. Furthermore, the film thickness of the third region **63** reaches the maximum film thickness $T3_{max}$, and then, gradually decreases in the direction toward the tip end side of the exposed end portion **44** to reach the predetermined film thickness $T3_{mid}$.

[0092] In addition, the minimum film thickness $T3_{min}$ of the third region **63** is equal to the maximum film thickness $T2_{max}$ of the second region **62** ($T3_{min}=T2_{max}$). Also, the maximum film thickness $T3_{max}$ of the third region **63** is larger than the maximum film thickness $T2_{max}$ of the second region **62** ($T3_{max}>T2_{max}$). Furthermore, the predetermined film thickness $T3_{mid}$ is smaller than the maximum film thickness $T3_{max}$ and is larger than the minimum film thickness $T3_{min}$ ($T3_{min}<T3_{mid}<T3_{max}$).

[0093] Thus, in the third step, the pull-up speed that is gradually increased to the pull-up maximum speed $V3$ in the second step is continuously gradually increased to a pull-up maximum speed $V4$ in the third step. Next, the pull-up speed gradually increased to the pull-up maximum speed $V4$ is gradually decreased to a predetermined speed $V5$.

[0094] More specifically, in the third step, the exposed end portion **44** of the core **41** is vertically pulled up from the solution **70** at a pull-up speed satisfying a quadratic equation ($speed=ix\cdot sup.2+jx+k$ (terms “ i ”, “ j ”, and “ k ” are rational numbers)).

[0095] As described above, the pull-up maximum speed $V3$ in the second step is also the pull-up

start speed in the third step. In other words, the pull-up start speed in the third step is equal to or higher than the pull-up maximum speed **V3** in the second step.

[0096] Also, the predetermined speed **V5** is lower than the pull-up maximum speed **V4** in the third step and is higher than the pull-up start speed **V3** in the third step (pull-up maximum speed in the second step) ($V3 < V5 < V4$).

Fourth Step and Fifth Step

[0097] Although not illustrated, the fourth step is a step of forming the fourth region **64** shown in FIG. 4, and the fifth step is a step of forming the fifth region **65** shown in FIG. 4.

[0098] As described above, in the entire fourth region **64**, the film thickness of the fourth region **64** is smaller than the film thickness of the first region **61**, and is constant.

[0099] Thus, in the fourth step, the pull-up speed is maintained at a constant speed lower than the pull-up start speed **V1** in the first step.

[0100] As described above, the fifth region **65** covers the entire tip end surface **44b** of the exposed end portion **44**, and has a substantially semispherical shape. Thus, in the fifth step, the pull-up speed is adjusted so as to form the fifth region **65** having the shape shown in FIG. 5.

[0101] As shown in FIG. 9D, at the time of the end of the pull-up step including the above-described first step to fifth step, the solution **70** having the desired thickness is adhered to the periphery of the exposed end portion **44**. Then, when the solution **70** is cured, the light-scattering resin film **5** is formed. Here, the cured solvent **71** becomes the base material **50** of the light-scattering resin film **5**.

[0102] Note that the viscosity of the silicone resin that is the solvent **71** of the solution **70** greatly changes depending on the temperature. Thus, in the pull-up step, the temperature of the solution **70** may be adjusted by heating means (for example, heater) arranged on the periphery of the container **72** so that the viscosity of the solution **70** becomes a desired viscosity. In the present embodiment, the temperature of the solution **70** is maintained at about 40° C. during the pull-up step.

[0103] In the present embodiment, after the pull-up step, heat treatment for cross-linking the solvent **71** (base material **50**) is performed.

<Light-Emission Intensity>

<<Measuring Method and Measuring Results>>

[0104] FIG. 11 is a graph showing measuring results of light-emission intensity of the peripheral light-emitting optical fiber **3** manufactured by the above-described manufacturing method.

[0105] In this measurement, laser light output from the light source **21** shown in FIG. 1 was made incident on the core **41** of the optical fiber **4** from the base end side of the peripheral light-emitting optical fiber **3**, and the intensity of light emitted in a radial direction from the light-scattering resin film **5** was measured.

[0106] For measurement of the light intensity, an optical power meter including a semiconductor sensor that generates a voltage in accordance with the intensity of the received light was used. Also, while the peripheral light-emitting optical fiber **3** is moved in the axial direction by a robot, the intensity of the light emitted from the light-scattering resin film **5** was measured at each 0.4-mm interval.

[0107] The vertical axis of the graph shown in FIG. 11 represents the light intensity (the output voltage of the semiconductor sensor included in the optical power meter), and the horizontal axis thereof represents a measuring position of the light intensity (a distance from the start edge of the first region **61** of the light-scattering resin film **5** to the measuring position).

[0108] In this measurement, the light intensity was measured to have a substantially same level in a section (hereinafter referred to as “light-emission intensity uniform section”) from a position away by 2.0 mm from the end surface of the cladding **42** to a position away by 52.0 mm therefrom.

[0109] More specifically, the light-emission intensity uniformity in the light-emission intensity uniform section was 80%. The light-emission intensity uniformity (%) is an index for indicating the uniformity of the light-emission intensity, and is a value obtained by dividing a minimum value of

the light-emission intensity measured in the light-emission intensity uniform section by a maximum value thereof and then multiplying the divided result by 100.

[0110] Also, the light-emission efficiency of the peripheral light-emitting optical fiber **3** was 88%. The light-emission efficiency (%) is an index for indicating the light utilization efficiency, and is a value obtained by dividing the intensity of light emitted from the light-scattering resin film **5** by the intensity of light made incident on the core **41** of the optical fiber **4** and then multiplying the divided result by 100.

[0111] From this measurement, it has been confirmed that the peripheral light-emitting optical fiber **3** manufactured by the above-described manufacturing method provides the substantially uniform light intensity distribution as a whole.

Operation and Effect of Present Embodiment

[0112] The peripheral light-emitting optical fiber **3** according to the present embodiment provides the substantially uniform light intensity distribution as a whole, because of balance between the decrease in the light intensity in the exposed end portion **44** of the core **41** and the increase in the light-scattering particles **51** contained in the light-scattering resin film **5** covering the exposed end portion **44** of the core **41**.

[0113] The light intensity in the exposed end portion **44** of the core **41** gradually decreases in a direction from the base end side toward the tip side of the exposed end portion **44**. On the other hand, the film thickness of the light-scattering resin film **5** gradually increases in a direction from the base end side to the tip side of the exposed end portion **44**.

[0114] Here, the solution **70** that is the material of the light-scattering resin film **5** is mixed so that the light-scattering particles **51** are uniformly dispersed. Therefore, when the film thicknesses of the first region **61**, the second region **62**, and the third region **63** of the light-scattering resin film **5** are equal to one another, the amounts (the number) of the light-scattering particles **51** included in the respective regions are substantially equal to one another.

[0115] In this case, by the gradual decrease in the light intensity in the exposed end portion **44**, the intensity of light emitted from the light-scattering resin film **5** is also gradually decreased. That is, the uniform light intensity distribution is not obtained.

[0116] However, in the peripheral light-emitting optical fiber **3** according to the present embodiment in which the film thickness of the light-scattering resin film **5** gradually increases in the direction from the base end side toward the tip side of the exposed end portion **44**, the second region **62** includes more light-scattering particles **51** than that in the first region **61**, and the third region **63** includes more light-scattering particles **51** than that in the second region **62**.

[0117] As a result, the decrease in the light intensity in the exposed end portion **44** is cancelled by the increase in the light diffuse reflection due to the light-scattering particles **51**, and the substantially uniform light intensity distribution is obtained as a whole.

[0118] Also, in the present embodiment, an annular small-thickness portion is provided closer to the tip end side than the third region **63** having the thickest portion of the light-scattering resin film **5**. From another viewpoint, the fourth region **64** positioned closer to the tip end side than the third region **63** has a constant smaller film thickness than that of the first region **61**.

[0119] The fourth region **64** (annular small-thickness portion) as described above is formed by the fourth step of the pull-up step in which the pull-up speed is maintained at a constant speed lower than the pull-up start speed **V1** in the first step.

[0120] By the fourth step having the lower pull-up speed than those of the first step to the third step, formation of a large resin ball on the periphery of the tip end surface **44b** of the exposed end portion **44** is prevented. If a large resin ball is formed on the periphery of the tip end surface **44b** of the exposed end portion **44**, the resin ball contains many light-scattering particles **51**. Thus, the light intensity in this resin ball is locally strengthened, and there is a risk of decrease in the light intensity distribution uniformity.

[0121] That is, the fourth region **64** (annular small-thickness portion) formed by the fourth step has

a function of suppressing the decrease in the light intensity distribution uniformity.
[0122] Furthermore, in the present embodiment, the film thickness of the third region **63** reaches the maximum film thickness, and then, gradually decreases. From another viewpoint, a gradually-decreasing portion having a gradually-smaller film thickness is formed beyond the thickest portion of the light-scattering resin film **5**. As a result, rapid decrease in the light intensity is suppressed, and the light intensity distribution uniformity is further improved. In other words, the gradually-decreasing portion is an extra length portion for suppressing the rapid decrease in the light intensity due to the influence of the annular small-thickness portion.

[0123] The present invention is not limited to the above-described embodiments, and various modifications can be made within the scope of the present invention. For example, the film thickness of the light-scattering resin film **5** in each region can be appropriately changed on condition that a predetermined relation is satisfied. Nevertheless, if the peripheral light-emitting linear light guide member **3** is used as the catheter, the tip portion of the peripheral light-emitting linear light guide member **3** to be inserted into the body of the patient **P** is desirably as thin as possible. From this viewpoint, the total of the maximum film thickness $T1_{max}$ of the first region **61**, the maximum film thickness $T2_{max}$ of the second region **62**, and the maximum film thickness $T3_{max}$ of the third region **63** is desirably smaller than the diameter $D1$ of the core **41** ($T1_{max}+T2_{max}+T3_{max}<D1$).

Claims

1. A peripheral light-emitting linear light guide member comprising: an optical fiber including a cladding and a core exposed from the cladding at one end portion; and a light-scattering resin film including a light-transmittable base material having a refractive index higher than that of the core and light-scattering particles dispersed in the base material, wherein the light-scattering resin film includes a first region, a second region, and a third region covering an outer periphery surface of the end portion of the core, the first region, the second region, and the third region are aligned in this order from a base end side to a tip end side of the end portion, a minimum film thickness of the second region is equal to or larger than a maximum film thickness of the first region, a minimum film thickness of the third region is equal to or larger than a maximum film thickness of the second region, each film thickness of the first region and the second region gradually increases in a direction toward the tip end side of the end portion, and reaches the maximum film thickness of each region, and a film thickness of the third region gradually increases in the direction toward the tip end side of the end portion, and reaches a maximum film thickness of the third region, and then, gradually decreases to a predetermined film thickness.
2. The peripheral light-emitting linear light guide member according to claim 1, wherein the predetermined film thickness is smaller than the maximum film thickness of the third region and larger than the minimum film thickness of the third region.
3. The peripheral light-emitting linear light guide member according to claim 1, wherein the light-scattering resin film further includes a fourth region positioned closer to the tip end side of the end portion than the third region and covering the outer periphery surface of the end portion and a fifth region positioned closer to the tip end side of the end portion than the fourth region and covering an end surface of the end portion, the fourth region has a film thickness smaller than the film thickness of the first region over the entire fourth region, and a step surface parallel to the end surface of the end portion of the core is provided between the third region and the fourth region.
4. The peripheral light-emitting linear light guide member according to claim 1, wherein a length of the second region is equal to or larger than a length of the first region, and a length of the third region is smaller than the length of the first region and smaller than the length of the second region.
5. A method of manufacturing a peripheral light-emitting linear light guide member, the peripheral light-emitting linear light guide member including: an optical fiber including a core exposed from a

cladding at one end portion; and a light-scattering resin film including a first region, a second region, and a third region covering an outer periphery surface of the end portion of the core, the method comprising: an optical-fiber processing step of exposing the end portion of the core from the cladding; a film-forming material preparing step of preparing a solution containing light-scattering particles dispersed in a light-transmittable solvent having a refractive index higher than a refractive index of the core; an immersing step of immersing the end portion of the core into the solution; and a pull-up step of pulling up the end portion of the core from the solution, wherein the pull-up step includes at least a first step of forming the first region, a second step of forming the second region, and a third step of forming the third region, the first step, the second step, and the third step are performed in this order, a pull-up start speed in the second step is equal to or higher than a pull-up maximum speed in the first step, a pull-up start speed in the third step is equal to or higher than a pull-up maximum speed in the second step, a pull-up speed in the first step is gradually increased from a pull-up start speed to the pull-up maximum speed in the first step, the pull-up speed in the second step is gradually increased from the pull-up start speed to the pull-up maximum speed in the second step, and the pull-up speed in the third step is gradually increased from the pull-up start speed to a pull-up maximum speed in the third step, and then, is gradually decreased to a predetermined speed.

6. The method of manufacturing the peripheral light-emitting linear light guide member according to claim 5, wherein the predetermined speed is lower than the pull-up maximum speed in the third step and higher than the pull-up start speed in the third step.

7. The method of manufacturing the peripheral light-emitting linear light guide member according to claim 5, wherein the light-scattering resin film further includes a fourth region covering the outer periphery surface of the end portion and a fifth region covering an end surface of the end portion, the pull-up step further includes a fourth step of forming the fourth region to be performed after the third step and a fifth step of forming the fifth region to be performed after the fourth step, and a pull-up speed in the fourth step is a constant speed lower than the pull-up start speed in the first step.

8. The method of manufacturing the peripheral light-emitting linear light guide member according to claim 5, wherein the pull-up speed in the first step satisfies a cubic equation ($\text{speed} = ax.\text{sup.3} + bx.\text{sup.2} + cx + d$ (terms “a”, “b”, “c”, and “d” are rational numbers)), the pull-up speed in the second step satisfies a cubic equation ($\text{speed} = ex.\text{sup.3} + fx.\text{sup.2} + gx + h$ (terms “e”, “f”, “g”, and “h” are rational numbers)), and the pull-up speed in the third step satisfies a quadratic equation ($\text{speed} = ix.\text{sup.2} + jx + k$ (terms “i”, “j”, and “k” are rational numbers))).
